

Exercise 1.1

1. If $A = \begin{pmatrix} -5 & 6 & 1 \\ 2 & -2 & 9 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 8 & -4 \\ 1 & 2 & -9 \end{pmatrix}$, then find the matrices $7A$, $3A + 2B$, and $2A - 3B$.
2. If $A = \begin{pmatrix} -1 & 0 & 6 \\ 3 & 8 & 9 \\ -3 & 4 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 2 & -7 \\ -1 & -5 & 2 \\ 4 & 0 & 1 \end{pmatrix}$, then find the matrices $-2A + 4B$, and $5A - 3B$.
3. If $A = \begin{pmatrix} -1 & 4 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 0 \\ -4 & 5 \end{pmatrix}$, compute AB and BA .
4. If $A = \begin{pmatrix} 6 & 0 \\ -4 & -2 \\ 9 & 8 \end{pmatrix}$ and $B = \begin{pmatrix} 8 & 5 \\ -2 & -3 \\ 0 & 3 \end{pmatrix}$, find AB and BA (if possible).
5. If $A = \begin{pmatrix} -1 & 4 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 0 \\ 6 & 5 \end{pmatrix}$, then prove that $(AB)^T = B^T A^T$.
6. If $A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 0 & 5 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & -2 & 8 \\ 0 & 6 & -2 \\ -1 & 0 & 9 \end{pmatrix}$ then prove that $(AB)^T = B^T A^T$.

Exercise 1.2

1. Find the value of the determinant $\begin{vmatrix} 7 & -3 & 0 \\ 3 & 4 & 3 \\ -5 & 0 & -5 \end{vmatrix}$.
2. Find the values of the constant α so that the determinant of the following matrix becomes zero

$$A = \begin{pmatrix} 2 & 1 & \alpha \\ 0 & 3 & 4 \\ 1 & -\alpha & 1 \end{pmatrix}.$$

3. Find the value of the determinant $\begin{vmatrix} 3 & -1 & 5 \\ -3 & 4 & -3 \\ 5 & -1 & 0 \end{vmatrix}$.

Exercise 1.3

1. Find the inverse of the following matrices (if possible) using cofactor method and elementary row operations (ERO) method, also justify your answers.

$$a. A = \begin{pmatrix} 3 & 2 \\ 1 & -4 \end{pmatrix}, \quad b. A = \begin{pmatrix} 4 & -2 \\ 7 & 6 \end{pmatrix}, \quad c. A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & -1 & 0 \end{pmatrix},$$

$$d. B = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 1 & -1 & 1 \end{pmatrix}, \quad e. A = \begin{pmatrix} -2 & 3 & 1 \\ 4 & 5 & 8 \\ 9 & -1 & 1 \end{pmatrix}, \quad f. A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix},$$

2. If $A = \begin{pmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{pmatrix}$, find AB , BA , $A^{-1}B$, $B^{-1}A$ and $(AB)^{-1}$.

3. If $A = \begin{pmatrix} -1 & -8 & -3 \\ 2 & 0 & 5 \\ 3 & 6 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$, find $A^{-1}B$.

Exercise 1.4

1. Reduce the following matrices to echelon form and hence, find the rank of the following matrices:

$$M = \begin{pmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 3 & 2 \\ 5 & 9 & 12 & 14 \end{pmatrix} \text{ and } P = \begin{pmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{pmatrix}.$$

Ans: 2

Ans: 2

Ans: 2

2. Find the rank of the following matrices:

$$\text{i) } A = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & 1 \\ 0 & 3 & 0 \end{pmatrix}, \quad \text{ii) } A = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 4 & 2 \\ -1 & 2 & 6 \end{pmatrix}, \quad \text{iii) } A = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \end{pmatrix}.$$

Ans: 2

Ans: 2

Ans: 3

References:

- ❑ Linear Algebra and its applications by David C Lay, Pearson, 4th Edition.
- ❑ Linear Algebra and its applications by Lay and Lay, Pearson, 5th Edition

Exercise 2

1. Find the rank of the matrices corresponding to each of the following systems of linear equations. Hence test whether the systems of linear equations are consistent. If consistent find the solutions of the system. Also check your answer by direct substitution.

- | | | |
|--|--|---|
| <p>a. $2x_1 - 3x_2 = -2$
 $2x_1 + x_2 = 1$
 $3x_1 + 2x_2 = 1$
 Ans: Inconsistent</p> | <p>b. $x_2 - 4x_3 = 8$
 $2x_1 - 3x_2 + 2x_3 = 1$
 $5x_1 - 8x_2 + 7x_3 = 1$
 Ans: Inconsistent</p> | <p>c. $x - 2y + z = 0$
 $2y - 8z = 8$
 $-4x + 5y + 9z = -9$
 Ans. (29, 16, 3)</p> |
| <p>d. $x + y + 2z = 0$
 $y + z = 0$
 $-2x + 3y + z = 0$
 Ans. $(-a, -a, a)$</p> | <p>e. $x + 2y + 4z = -13$
 $3x - y + z = 5$
 $x + 3y = 3$
 Ans. (3, 0, -4)</p> | <p>f. $x_1 - x_2 + 2x_3 = 5$
 $2x_1 + x_2 - x_3 = 2$
 $2x_1 - x_2 - x_3 = 4$
 Ans: (2, -1, 1)</p> |
| <p>g. $2x - y - 5z = 4$
 $x + y + z = -3$
 $-x - 4y + z = 4$
 Ans: $(-1, -1, -1)$</p> | <p>h. $-x + 2y + 2z = -2$
 $3x + 2y - z = 9$
 $x + 4y + z = 5$
 Ans: (2, 1, -1)</p> | <p>i. $x_1 + 2x_2 - 2x_3 = 2$
 $-x_1 + x_2 - 2x_3 = -1$
 $x_1 - 4x_2 - 2x_3 = 8$
 Ans: (2, -1, -1)</p> |
| <p>j. $2x - 3y + 4z = 8$
 $3x + 4y - 5z = -4$
 $5x - 7y + 6z = 9$
 Ans: (1, 2, 3)</p> | <p>k. $3x + y + 2z = 14$
 $2y + 5z = 22$
 $2x + 5y - z = -22$
 Ans: (2, -4, 6)</p> | <p>l. $x + 2y - 3z = 6$
 $2x - y + 4z = 2$
 $4x + 3y - 2z = 14$
 Ans: $(2 - a, 2 + 2a, a)$</p> |
| <p>m. $-x + y - z = 0$
 $3x - y + 2z = -2$
 $2x + 4y + 3z = 2$
 Ans: $(-2, 0, 2)$</p> | <p>n. $x + 2y + 3z = 1$
 $x + 3y + 6z = 3$
 $2x + 6y + 13z = 5$
 Ans: $(-6, 5, -1)$</p> | <p>o. $x + y + z = -1$
 $x - y + z = -5$
 $2x + y - z = 5$
 Ans: (0, 2, -3)</p> |
| <p>p. $x + 2y + 3z = 5$
 $2x + 5y + 3z = 3$
 $x + 8z = 17$
 Ans: (1, -1, 2)</p> | <p>q. $3x + 2y - z = -15$
 $5x + 3y + 2z = 0$
 $3x + y + 3z = 11$
 $-6x - 4y + 2z = 30$
 Ans: $(-4, 2, 7)$</p> | |

2. Solve the following system of linear equations by using Cramer's rule and Matrix inversion methods (If Applicable).

- | | | |
|--|---|---|
| <p>a. $-3x + 2y - 3z = -8$
 $2x - y + z = 4$
 $x + 2y - 4z = -2$
 Ans: (2, 2, 2)</p> | <p>b. $x_1 + x_2 + 2x_3 = 8$
 $-x_1 - 2x_2 + 3x_3 = 1$
 $3x_1 - 7x_2 + 4x_3 = 10$
 Ans: (3, 1, 2)</p> | <p>c. $3x - y + z = -2$
 $x + 5y + 2z = 6$
 $2x + 3y + z = 0$
 Ans: $(-2, 0, 4)$</p> |
|--|---|---|

d.	$x_1 + 2x_2 + x_3 = 2$	e.	$x + y + 2z = 1$	f.	$2x_1 - x_2 - x_3 = 6$
	$2x_1 - x_2 + 2x_3 = -1$		$y + z = 1$		$x_1 + 3x_2 + 2x_3 = 1$
	$3x_1 - 4x_2 - 3x_3 = -16$		$-2x + 3y + z = 3$		$3x_1 - x_2 - 5x_3 = 1$
	Ans: $(-2, 1, 2)$		Ans: Determinant zero, not applicable		Ans: $(3, -2, 2)$

3. Determine the value(s) of λ such that the following system of linear equations has (i) no solution, (ii) more than one solution, and (iii) a unique solution.

$$\begin{cases} \lambda x + y + z = 1 \\ x + \lambda y + z = 1. \\ x + y + \lambda z = 1 \end{cases}$$

Ans: (i) $\lambda = -2$, (ii) $\lambda = 1$, (iii) $\lambda \neq 1, \lambda \neq -2$.

4. Determine the value(s) of λ such that the following system of linear equations has (i) no solution, (ii) more than one solution, and (iii) a unique solution.

$$\begin{cases} x - 3z = -3 \\ 2x + \lambda y - z = -2. \\ x + 2y + \lambda z = 1 \end{cases}$$

Ans: (i) $\lambda = -5$, (ii) $\lambda = 2$, (iii) $\lambda \neq 2, \lambda \neq -5$.

5. Determine the value(s) of λ and μ such that the following system of linear equations has (i) no solution, (ii) more than one solutions and (iii) a unique solution.

$$\begin{cases} x + y + z = 2 \\ x + 3y + \lambda z = 6. \\ x + 2y + 3z = \mu \end{cases}$$

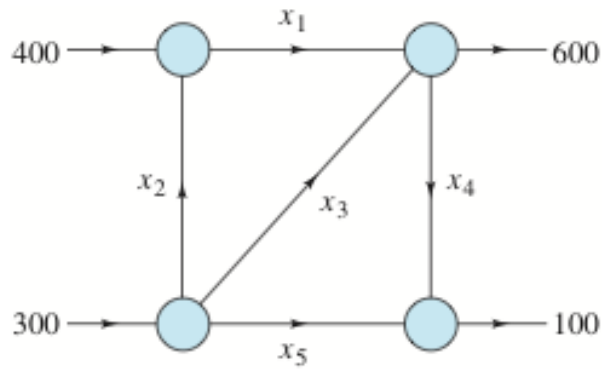
Ans: (i) $\lambda = 5, \mu \neq 4$, (ii) $\lambda = 5, \mu = 4$, (iii) $\lambda \neq 5$.

6. Determine the value(s) of λ such that the following system of linear equations has (i) no solution, (ii) more than one solution, and (iii) a unique solution.

$$\begin{cases} x + y + \lambda z = 1 \\ x + \lambda y + z = \lambda. \\ \lambda x + y + z = \lambda^2 \end{cases}$$

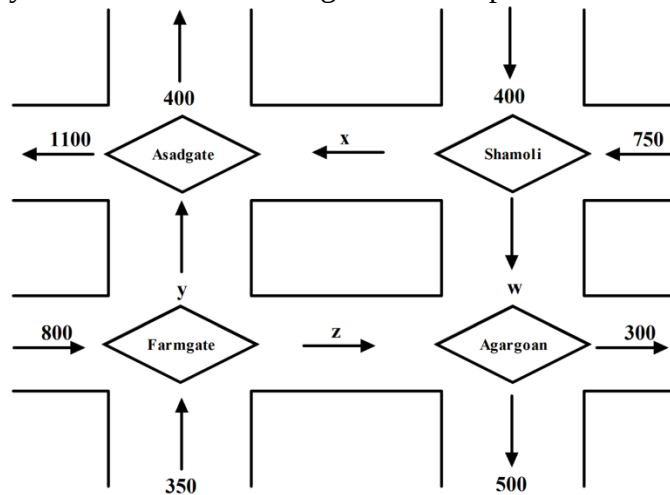
Ans: (i) $\lambda = -2$, (ii) $\lambda = 1$, (iii) $\lambda \neq 1, \lambda \neq -2$.

7. The network in the figure below shows the traffic flow (in vehicles per hour) over the several one-way streets. Determine the general flow pattern for the network.

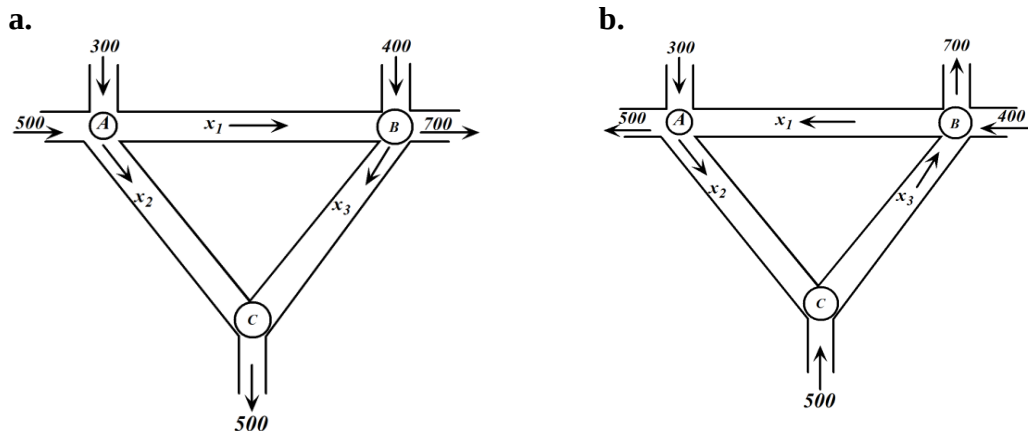


Ans: Here x_3 and x_5 are free variables.

8. The network in the figure below shows the traffic flow (in vehicles per hour) over the several one-way streets. Determine the general flow pattern for the network.



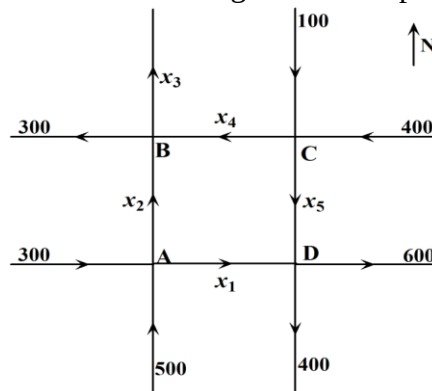
9. The network in the figure below shows the traffic flow (in vehicles per hour) over the several one-way streets. Determine the general flow pattern for the network.



Ans: (a) Here x_3 is the free variable. The system has an infinite number of solutions. But to remove negativity x_3 must be between 0 to 500.

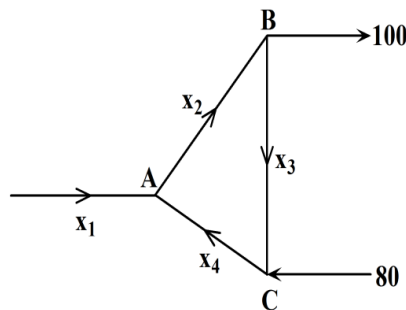
(b) Here x_3 is the free variable. The system has an infinite number of solutions. But to remove negativity x_3 must be greater than 500.

10. The network in the figure below shows the traffic flow (in vehicles per hour) over the several one-way streets. Determine the general flow pattern for the network.



Ans: Here x_5 is the free variable. The system has an infinite number of solutions. But to remove negativity x_5 must start from 200.

11. Find the general flow pattern of the network system in the figure. Assuming that the flows are all nonnegative, what is the smallest possible value for x_4 .

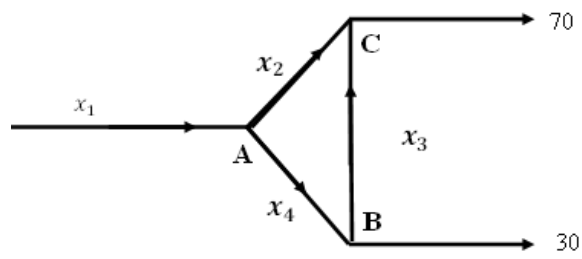


12. A company produces three products x, y, z every day. Their total production on a certain day is 45 tons. It is found that the production of z exceeds the production of x by 8 tons while the total production of ' x ' and ' z ' is twice the production of ' y '. Determine the production level of each product.

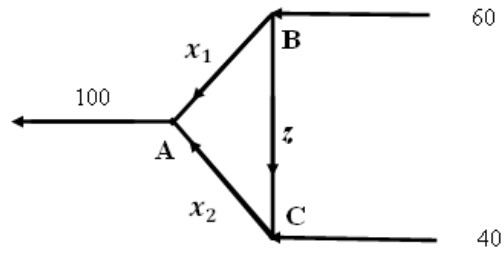
Ans: $x = 1$; $y = 15$; $z = 19$.

13. Construct the system of linear equations from the following diagrams, reduced the system to echelon form and finally find the general flow pattern, where x_i is the number of cars.

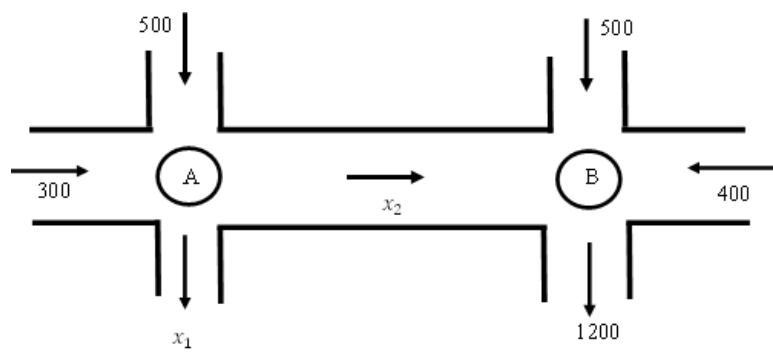
(i)



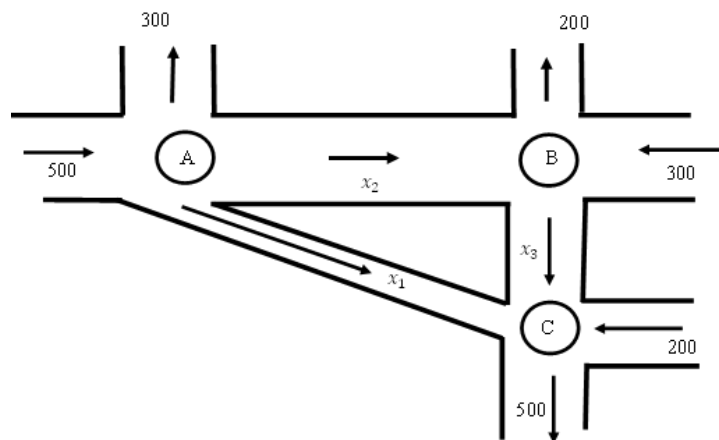
(ii)



(iii)

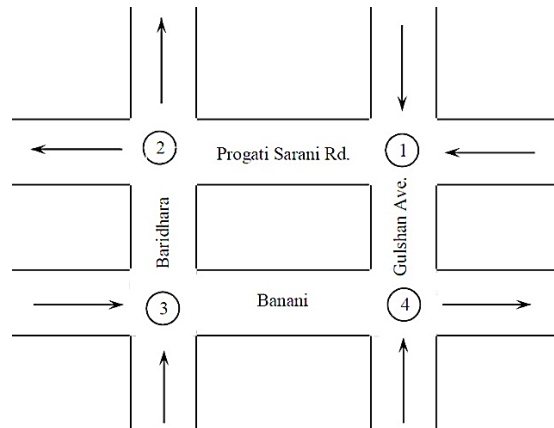


(iv)



14. Traffic congestion is encountered at the intersections shown in the figure. All the streets are one-way and in the directions shown. In order for effective movement of traffic, it is necessary that for every car that arrives at a given corner, another car must leave so that the number of cars arriving per unit time must equal the number of cars leaving per unit time. Traffic engineers gather the following information:

- 600 cars per hour come down Progati Sarani Rd. to intersection #1 and 300 cars per hour enter intersection #1 on Gulshan Ave.
- 650 cars per hour leave intersection #2 along Progati Sarani Rd. and 50 cars per hour leave intersection #2 along Baridhara.
- 350 cars per hour come up Banani to intersection #3 and 50 cars per hour enter intersection #3 along Baridhara.
- 400 cars per hour leave intersection #4 along Banani and 300 cars per hour enter intersection #4 from Gulshan Ave.



Find n_1 , n_2 , n_3 , and n_4 , where n_1 denotes the number of cars traveling per hour along Progati Sarani Rd. from intersection #1 to intersection #2, n_2 denotes the number of cars traveling per hour along Baridhara from intersection #3 to intersection #2, n_3 denotes the number of cars traveling per hour along Banani from intersection #3 to intersection #4, and n_4 denotes the number of cars traveling per hour along Gulshan Ave. from intersection #1 to intersection #4.

15. Encode the message **TRY YOUR BEST** by using matrix, $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. Also decode your encrypted message using the same matrix.

16. Encode the message **HONESTY IS THE BEST POLICY** by using matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ given above.

17. Encrypt the message **EFFORT NEVER DIES** by the provided matrix, $B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.

18. The encoded message is 28 13 28 20 23 5. Decode this message by using matrix, $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Linear Programming:

Reference: Operation Research by Hamdy A. Taha, 10th edition.

Problem-2.1-1 and 2.1-2 (Page-45-52), problem 2.7, 2.8, 2.9, 2.13 (Page-77,78).

Exercise 3.1

1. Determine the period of the following functions.

(a) $f(x) = \sin 5x$, (b) $f(x) = \tan 7x$, (c) $f(x) = \cos^2 x$,

(d) $f(x) = \sin^2 x - \cos^2 x$,

(e) $f(x) = \sin 2x + \cos 3x$.

(a) Ans: $\frac{2\pi}{5}$. (b) Ans: $\pi/7$. (c) Ans: π . (d) Ans: π . (e) Ans: 2π .

2. Determine whether the following functions are even, odd or neither.

(a) $f(x) = \sin x + \cos\left(\frac{x}{3}\right)$, (b) $f(x) = x^3 - x$, (c) $f(x) = \tan x + \sin 2x$,

(d) $f(x) = e^{x^2}$,

(e) $f(x) = \cos x$, (f) $f(x) = \frac{e^{3x} + e^{-3x}}{2}$, (g) $f(x) = \frac{e^x - e^{-x}}{2}$, (h) $f(x) = x^3$.

Exercise 3.2

1. Sketch the graph and find the Fourier coefficients and then Fourier series of the function of $f(x) = x^2$ in the interval $-\pi \leq x \leq \pi$.

Ans: $a_0 = \frac{2\pi^2}{3}$, $a_n = \frac{4(-1)^n}{n^2}$, $b_n = 0$.

2. Sketch the graph and find the Fourier coefficients and then Fourier series of the function of $f(x) = 2x$ in the interval $-2 \leq x \leq 2$.

Ans: $a_0 = 0$, $a_n = 0$, $b_n = -\frac{8(-1)^n}{n\pi}$.

3. Sketch the graph and obtain the Fourier series of the function $f(x) = \begin{cases} 1 & \text{when } -\pi < x < 0 \\ 2 & \text{when } 0 < x < \pi \end{cases}$.

Ans: $a_0 = 3$, $a_n = 0$, $b_n = -\frac{1}{\pi n}[(-1)^n - 1] = \begin{cases} 0 & \text{when } n = \text{even} \\ \frac{2}{\pi n} & \text{when } n = \text{odd} \end{cases}$.

4. Sketch the graph and obtain the Fourier series of the function $f(x) = |x|$ in the interval $-\pi \leq x \leq \pi$.

Ans: $a_0 = \pi$, $a_n = \begin{cases} \frac{-4}{\pi n^2} & \text{when } n = 1, 3, 5, \dots \\ 0 & \text{when } n = 2, 4, 6, \dots \end{cases}$, $b_n = 0$.

5. Find the complex form of Fourier series of the following function:

- (i) $f(x) = 2x$ in the region, $-\pi \leq x \leq \pi$.
- (ii) $f(x) = x^2$ in the region, $\pi \leq x \leq \pi$.
- (iii) $f(x) = x^2 - x$ in the region, $-\pi \leq x \leq \pi$.

Exercise: 3.3

1. Sketch the odd and even extension and express $f(x) = x$ as a half range Fourier sine and cosine series in the interval $0 < x < 2$.

$$\text{Ans: } f_s(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi}{2}x\right), \quad f_c(x) = 1 + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{\{(-1)^n - 1\}}{n^2} \cos\left(\frac{n\pi}{2}x\right)$$

2. Sketch the odd and even extension and express $f(x) = x^2$ as a half range Fourier sine and cosine series in the interval $0 < x < \pi$.

$$\text{Ans: } f_s(x) = \sum_{n=1}^{\infty} \frac{2}{\pi} \left[\frac{\pi^2}{n} (-1)^{n+1} + \frac{2}{n^3} \{(-1)^n - 1\} \right] \sin nx$$

$$\text{and } f_c(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^{n+1} \cos nx.$$

3. Sketch the odd and even extension and find the half range Fourier sine and cosine series of

$$f(x) = \begin{cases} x, & 0 < x < 4 \\ 8 - x, & 4 < x < 8. \end{cases}$$

$$\text{Ans: } f_s(x) = \frac{32}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \sin \frac{n\pi x}{8}, \text{ and } f_c(x) = \frac{16}{\pi^2} \sum_{n=1}^{\infty} \left[\frac{2 \cos \frac{n\pi}{2} - \cos n\pi - 1}{n^2} \right] \cos \frac{n\pi x}{8}.$$

4. Sketch the odd and even extension and find the half range Fourier sine and cosine series of

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 2, & 1 < x < 2. \end{cases}$$

$$\text{Ans: } f_s(x) = \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} - \frac{4}{n\pi} (-1)^n \right) \sin \frac{n\pi x}{2}, \quad f_c(x) = \frac{3}{2} - \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin \frac{n\pi}{2} \cos \frac{n\pi x}{2}.$$

5. Sketch and find the half range Fourier sine and cosine series of $f(x) = \pi - x$ in the interval $0 < x < \pi$.

$$\text{Ans: } f_s(x) = 2 \left[\frac{\sin x}{1} + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \dots \right],$$

$$\text{and } f_c(x) = \frac{\pi}{2} + \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right].$$

Exercise: 3.4

1. Find the Fourier integral of the function

$$f(x) = \begin{cases} 0 & \text{when } x < -1 \\ 1 - x & \text{when } -1 \leq x \leq 1. \\ 0 & \text{when } x \geq 1 \end{cases}$$

$$\text{Ans: } f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\frac{2}{\pi w} \sin w \cos wx + \frac{2}{\pi} \left(\frac{1}{w} \cos w - \frac{1}{w^2} \sin w \right) \sin wx \right] dw.$$

2. Find the Fourier sine and cosine integral of the function

$$f(x) = \begin{cases} x^2, & 0 < x < 2 \\ 0, & x > 2 \end{cases}.$$

3. Find the Fourier integral of the function

$$f(x) = \begin{cases} 0 & \text{when } x < 0 \\ \frac{1}{2} & \text{when } x = 0 \\ e^{-x} & \text{when } x > 0 \end{cases}.$$

$$\text{Ans: } f(x) = \frac{1}{\pi} \int_0^{\infty} \frac{\cos ux + u \sin ux}{1+u^2} du.$$

4. Find the Fourier integral of $f(x) = e^{-kx}$ when $x > 0$ and $f(-x) = -f(x)$ for

$$k > 0 \text{ and hence prove that } \int_0^{\infty} \frac{w \sin(wx)}{w^2 + k^2} dw = \frac{\pi}{2} e^{-kx}, k > 0.$$

5. Find the Fourier integral of $f(x) = e^{-kx}$ when $x > 0$ and $f(-x) = f(x)$ for

$$k > 0 \text{ and hence prove that } \int_0^{\infty} \frac{\cos(wx)}{w^2 + k^2} dw = \frac{\pi}{2k} e^{-kx}, k > 0.$$

Exercise: 3.5

Sketch the graph and then find the (a) finite Fourier sine transform, and (b) finite Fourier cosine transform of the following functions:

1. $f(x) = 2x$ where $0 < x < 4$.

$$\text{Ans: } f_s(n) = \frac{32}{n\pi} (-1)^{n+1}, f_c(n) = \frac{32}{n^2\pi^2} [(-1)^n - 1].$$

2. $f(x) = x^2$, $0 < x < b$.

$$\text{Ans: } f_s(n) = \begin{cases} \frac{2b^3}{n^3\pi^3} (\cos n\pi - 1) - \frac{b^3}{n\pi} \cos n\pi, & n = 1, 2, 3, \dots \\ \frac{b^3}{3}, & n = 0 \end{cases}, f_c(n) = \frac{2b^3}{n^2\pi^2} (\cos n\pi - 1).$$

3. Sketch the graph and then find the (a) infinite Fourier sine transform, and (b) infinite

$$\text{Fourier cosine transform of } f(x) = \begin{cases} 1 & \text{when } 0 \leq x < 1 \\ 0 & \text{when } x \geq 1 \end{cases}.$$

$$\text{Ans: } f_s(w) = \sqrt{\frac{2}{\pi}} \frac{1}{w} (1 - \cos w), \quad f_c(w) = \sqrt{\frac{2}{\pi}} \frac{\sin w}{w}.$$

4. Sketch the graph and then find the (a) infinite Fourier sine transform, and (b) infinite Fourier

$$\text{cosine transform of } f(x) = \begin{cases} x & \text{when } 0 < x < 1 \\ 2 - x & \text{when } 1 < x < 2 \\ 0 & \text{when } x > 2 \end{cases}.$$

$$\text{Ans: } f_s(n) = \sqrt{\frac{2}{\pi}} \frac{1}{w^2} (2 \sin w - \sin 2w), \quad f_c(n) = \frac{1}{w^2} \sqrt{\frac{2}{\pi}} (2 \cos w - \cos 2w - 1).$$

5. Sketch the graph and then find the (a) infinite Fourier sine transform, and (b) infinite Fourier cosine transform of $f(x) = e^{-x}$, $x \geq 0$.

$$\text{Ans: } f_s(w) = \sqrt{\frac{2}{\pi}} \frac{w}{w^2 + 1}, \quad f_c(w) = \sqrt{\frac{2}{\pi}} \frac{1}{w^2 + 1}.$$

Reference:

W.H. Press et.al., Numerical Recipes in Fortran 90, Cambridge University Press 1996.

- Advance Engineering Mathematics, 10th edition by Erwin Kreyszig, Herbert Kreyszig, Edward J. Norminton.

Exercise: 3.6

Use the finite Fourier transform to solve the following boundary value problems:

- (1) $\frac{\partial U}{\partial t} = 3 \frac{\partial^2 U}{\partial x^2}$ with $U(0, t) = U(2, t) = 0$ and $U(x, 0) = x$, where $t > 0$ and $0 < x < 2$.

$$\text{Ans: } U(x, t) = \sum_{n=1}^{\infty} \frac{4(-1)^{n+1}}{n\pi} \sin \frac{n\pi x}{2} e^{-\frac{3}{4}n^2\pi^2 t}.$$

- (2) $\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}$ with $U(0, t) = U(\pi, t) = 0$ and $U(x, 0) = 2x$, where $t > 0$ and $0 < x < \pi$.

$$\text{Ans: } U(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left(\frac{-2\pi}{n} \cos n\pi e^{-n^2 t} \right) \sin nx.$$

- (3) $\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}$ with $U(0, t) = U(6, t) = 0$ and

$$U(x, 0) = \begin{cases} 1, & 0 < x < 3 \\ 0, & 3 < x < 6 \end{cases} \quad \text{where } t > 0 \text{ and } 0 < x < 6.$$

$$\text{Ans: } U(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(1 - \cos \frac{n\pi}{2} \right) e^{-\frac{n^2\pi^2 t}{36}} \sin \frac{n\pi x}{6}.$$